

2026 JAXA Intern Theme 2

Questions (please answer).

• Theme 2: Prototype development of X-ray 3D visualization tools

3 01 ThreeJS only.

⚡ 02_VTK.js

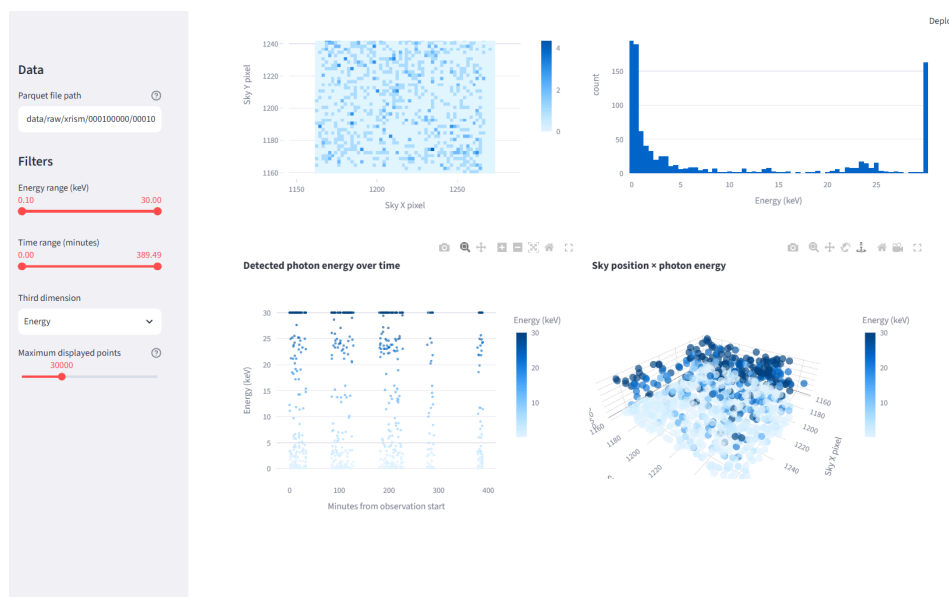
Full comparison

search+VTK current tool (for webapp).

test parquet file

Goal: Run the 3D tool locally through a web browser, display X-ray events from ASCA, Suzaku, Hitomi, and XRISM in 3D (users may select satellites/instruments), and facilitate the investigation of spatial variations in the energy spectra of extended sources, such as clusters of galaxies and supernova remnants.

Now Ken is able to run Ramon's tool :)



Today, I said I would concentrate on the 2D+Energy tool, giving up the 2D+Time representation.

However, having looked at Ramon's tool in which the 2D+Time representation is available, I now feel this is a nice capability to have. Let's keep it for now.

I would like to compare Plotly, Three.js, and PyVista (<https://pyvista.org/>)

We may look into this package: <https://pypi.org/project/astro-pyvista/>

Choice of framework

Comparative:

Framework	Data Representation	Memory Efficiency	Rendering / Interactivity	Web / DARTS Fit
Three.js (WebGL)	Sparse Point Clouds ($O(N)$)	Highest: Raw GPU vertex buffers; zero empty-bin allocation	60 FPS: Real-time GPU shader slicing and energy colormaps (LUT)	Native: Zero-install client-side web viewer (matches UDON3) PPTX
Plotly (Scatter3d)	Sparse Scatter Traces	Moderate: High JSON/ DOM overhead; lags past ~ 50k points	Moderate: Slicing requires complete trace re-renders	Easy for prototyping, but heavy memory footprint per session
PyVista / VTK	Continuous Meshes & Grids	Poor for sparse points: Optimized for dense volumetric grids	High (Native): Fast locally; sluggish over browser streams	Complex: Requires server-side rendering pipelines (trame)
CoDim	Proprietary CAD / Simulation	Unknown: Requires external cloud SaaS ingestion	High: Cloud-rendered	Incompatible: Closed-source, commercial SaaS

(`astro-pyvista` (PyPI) is a 2D optical/spectroscopic data reduction tool derived from 1980s `xvista` and cannot render 3D scenes.)

big thing is, we want a web app, so everything other than three.js cannot be a web app
so

option 1: use only three.js

⇒ BUT pretty slow / consuming

option 2:

1. When JAXA archives a dataset (e.g. `100011011_resolve`), the server runs Python **once** to calculate the physical celestial coordinates (RA, DEC, keV, minutes).
2. When a public user visits the website, their browser downloads the pre-calibrated data and displays the 3D cube instantly. (via three.js for ex)

⇒ BUT need action server side (⇒ costs?) + is okay to maintain in the future ?

option 3: API calls

1. If a researcher visits the website and enters custom target coordinates, the browser sends a quick background request to the server.
2. The server runs Python for 5 milliseconds and sends the projected coordinates back to the browser.

option 4: HYBRID ARCHITECTURE (RECOMMENDED)

1. Precompute standard columns (Energy, Time) in Parquet (Option A)
2. Do Tangential WCS Math directly in JavaScript in the browser!
 - Tangential projection (TAN) is just basic trigonometry (sin, cos)
 - Three.js can re-center on any RA/DEC instantly without a server

I also played a bit with Three.js (Ken)

[attachment:9c204ceb-60fb-406e-9a73-d8ff167a0f2c:index1.html](#)

For Three.js ⇒ see Ming's `xrism3d_tool` ⇒ pretty good for now, need pgrade but good basis

my (Théo) observations when testing it:

xrism3D by Ming

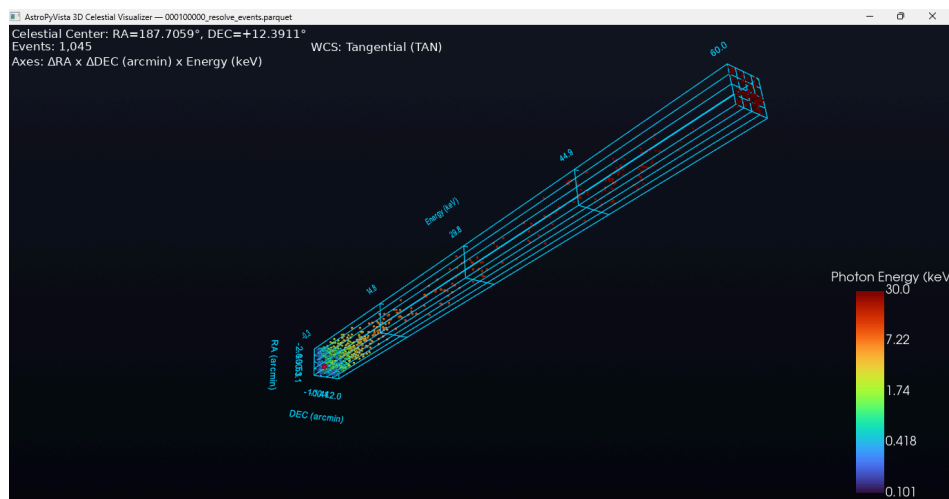
ok for the show of data
my laptop temp run high when runing the program
but wow no real UX, I need to see someone use that

infinite zoom when you touch a curve => no no
no explanations of what we're looking at => probaby ok for trained / scientists using the too
ls

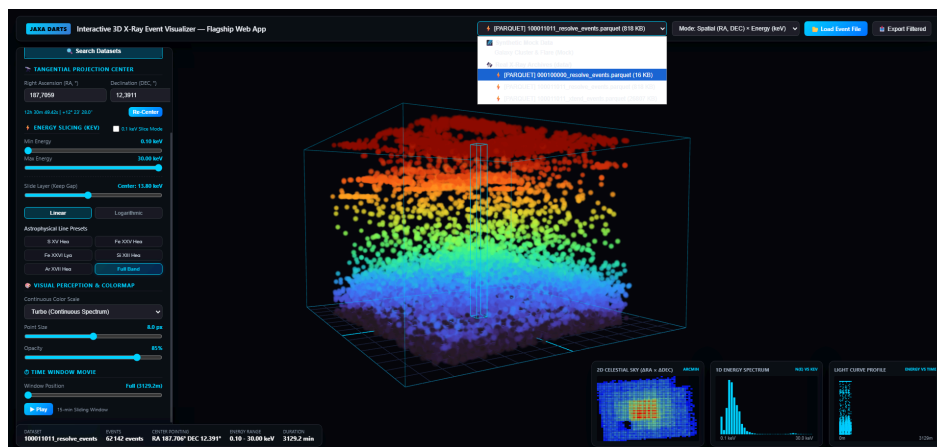
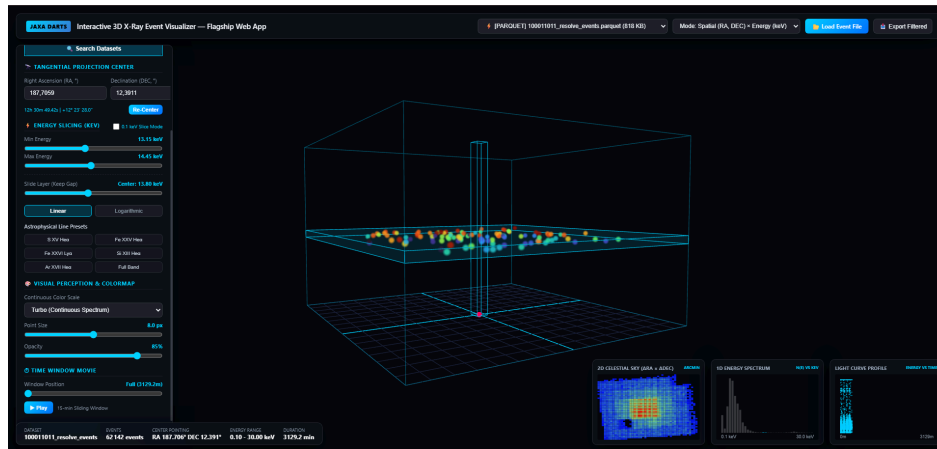
no all graphs are related directly, if i zoom on one the other don't change

about the 3D cube: => need to see it used for real not just to show how it work
nice for the energy filtering (but useful? to have a gauge ?? (slider) i guess)
nice choice for emission line , maybe useful to be able to select several (or not ?)
numbers and axis overstep on the datapoints, really useful to zoom that much ?

Astropyvista: (uses pyvista and numpy)



With Three.js with Python processing server side



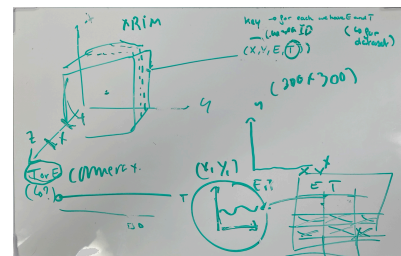
Analyse data visualize

1) Dataset $\{x, y\}$
 $\{RA, DEC\}$ as key
 Camera Id $\{Energy, Time\}$
 $\{XRAY\} \{E, Time\}$

3D Plot like Hough

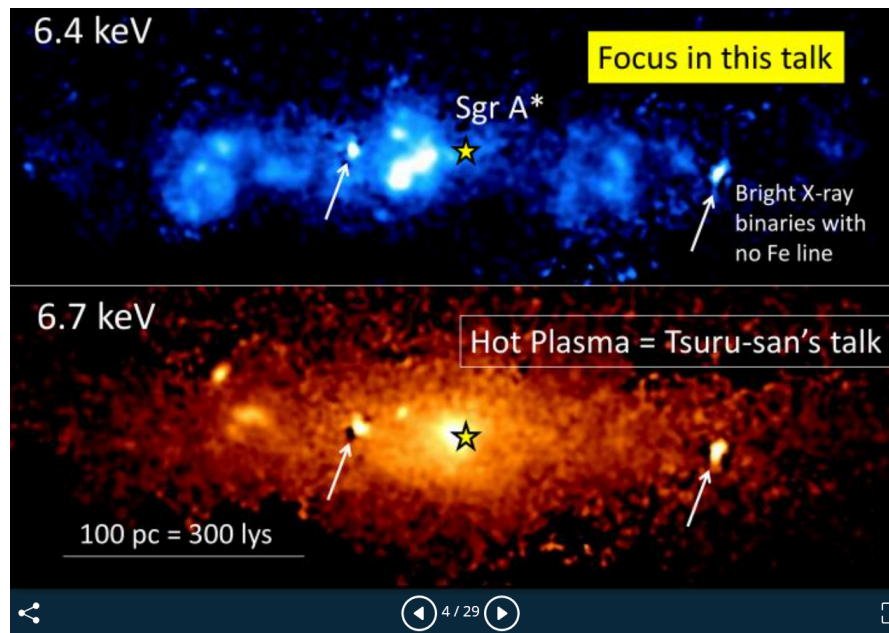
To add the layer energy
 To add the time series
 To select point in 3D
 $\rightarrow$ appears the 2^d time
 $\rightarrow$ Multiple points

Multiple packet (Multiple image)



Our Galactic center is a good example of indicating spatial variation of energy spectra

https://www-cr.scphys.kyoto-u.ac.jp/research/xray/xray-e_large.html



<https://www.slideserve.com/yule/suzaku-study-of-x-ray-emission-from-the-molecular-clouds-in-the-galactic-center>
<https://heasarc.gsfc.nasa.gov/docs/suzaku/gallery/science/gc.html>

Let's focus on the Galactic center region (Sgr A*). Position of the Galactic center is (0,0) in the Galactic coordinate, and it is (RA, DEC)=(266.404996, -28.936172). Images of 6.4 keV (neutral iron) and 6.7 keV (H-like ionized iron) are very different.

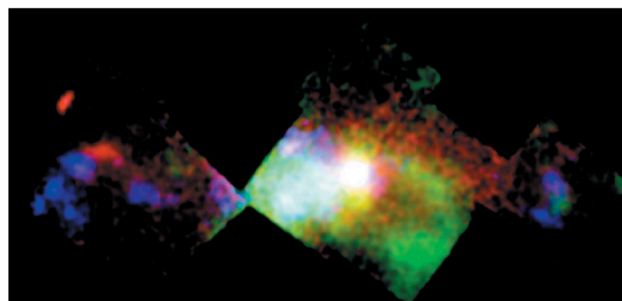


Fig. 2. The 3-color map of the Galactic center. Green = 2.45 keV (He like S $K\alpha$), Red = 6.7 keV (He like Fe $K\alpha$), and Blue = 6.4 keV (Neutral Fe $K\alpha$).

<https://academic.oup.com/ptps/article/doi/10.1143/PTPS.169.103/1876995>

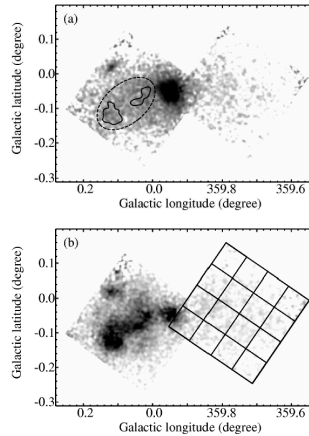


Fig. 1. Mosaic (2-pointings of the east and west fields) maps of the 6.7 keV line (a) and of the 6.4 keV line (b) bands. The east and west fields are the left (east), and the right (west) sides, respectively. Solid polygons in figure (a) are 6.4 keV clumps (here, the west clump is source 1, and the east clump is source 2), while the dashed ellipse is the background region. The background data for the 6.4 keV clumps were obtained by excluding the data of source 1 and 2. The grid in the west field shows 16 segmentations for the spatial resolved study (see text).

[0804.3309.pdf](#)

I imagine a following use-case:

1. User specify (RA, DEC)=(266.404996, -28.936172) (Galactic center) and Radius= 1 deg.
2. User may choose which satellites and instrument to choose.
 - a. Individual satellite
 - b. Individual instrument
 - c. Individual observation
3. The system searches for the packet files which is included in the Galactic center region with 1 deg radius of the selected satellites/instruments.
4. The system makes a new 3D array combining all the selected packets; projection on the new tangential plane at the (266.404996, -28.936172).
5. The 3D data are displayed in a browser, and users may manipulate with mouse.
6. User specifies, e.g., a 0.1 width-energy-slice. User can move the slice in the Energy axis direction, to investigate spatial variation of the energy spectra. Easily study spectral variation as shown in the pictures above.
7. (added on Aug 25) Users can bin and smooth the image (RA and DEC). Note, all the pretty picture above are properly smoothed. Users can bin the energy (spectrum). It is common to bin the spectrum, NOT common to smooth the spectrum. Still, it might be interesting to see how the spectral smoothing looks like.

Let's try two frameworks: PyVista (Ramon) and Three.js (Theo). Please leave your programs/scripts to [github](#), that would be a good resource for us (DARTS developers).

Please try to develop a prototype tool using either PyVista or Three.js. I will be the first user, and try the tasks 1 to 7 above. Note, selecting the packet files for given position and radius does not depend on the 3D framework (PyVista or Three.js), so this can be common.

Library for js:

<https://kitware.github.io/vtk-js/docs/>

VTk-js also sounds interesting, but we cannot do everything during this internship. I would like to know more about Three.js and PyVista. Let's see how far we can go with these two environments for the GC quick analysis task above (1 to 6). (Ken)

I forgot to tell you one thing important. PI 2 Energy (keV) conversion factor is different for different instruments:

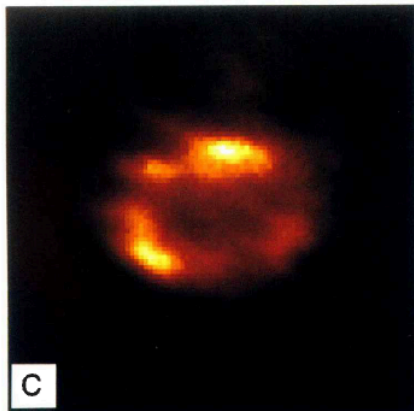
```
export const ENERGY_CONFIGS: Record<string, MissionEnergyConfig> = {
  xris: {
    resolve: {
      piToKeVFactor: 0.0005, // pi / 2000
      displayRange: { min: 0.3, max: 12.0 }
    },
    xtend: {
      piToKeVFactor: 0.006, // (pi * 6) / 1000
      displayRange: { min: 0.4, max: 12.0 }
    }
  },
  hitomi: {
    sxs: {
      piToKeVFactor: 0.0005,
      displayRange: { min: 0.3, max: 12.0 }
    },
    sxi: {
      piToKeVFactor: 0.006,
      displayRange: { min: 0.4, max: 15.0 }
    },
    hxi: {
      piToKeVFactor: 0.1,
      displayRange: { min: 3, max: 100.0 }
    }
  },
  suzaku: {
    xis: {
      piToKeVFactor: 0.00365,
      displayRange: { min: 0.2, max: 12.0 }
    }
  },
  asca: {
    sis: {
      piToKeVFactor: 0.0146,
      displayRange: { min: 0.4, max: 12.0 }
    },
    gis: {
      piToKeVFactor: 0.01178,
      displayRange: { min: 0.4, max: 12.0 }
    }
  }
};
```

Report 25/08/2026

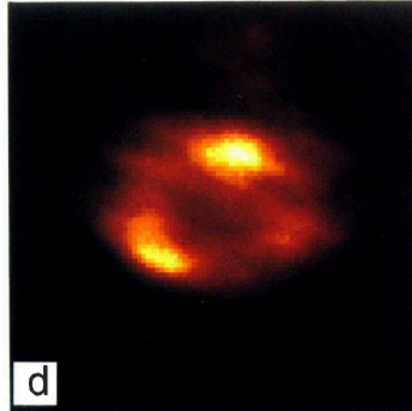
(2026/08/26 by Ken) Another good object to study spatial-spectral variation is super-nova remnants. Here is an example of Cas A by ASCA. The coordinate is (350.8584, 58.8113)

[0804.3309.pdf](#)

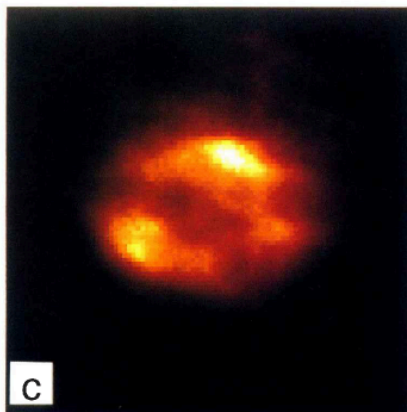
Si He $_{\alpha}$ Line Map Restored



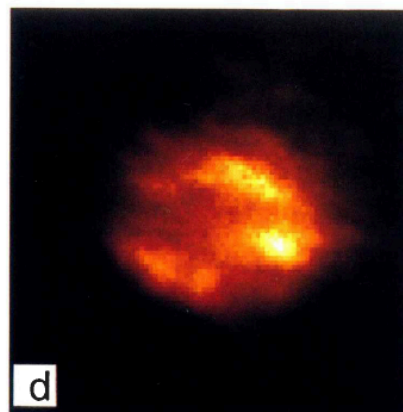
S He $_{\alpha}$ Line Map Restored



Low Energy
Continuum Restored



High Energy
Continuum Restored



Below, I attach the Cas A ASCA paper from which the picture above is taken. In this paper, they explain the Si He alpha line image is made from **1.75-1.95 keV**, and the S He alpha line image is made from **2.35 -2.52 keV**, and the low-energy continuum image is made from **1.55-1.75 keV**, and the high energy continuum image is made from **3.93-6.23 keV**.

[pasj_46_4_l151.pdf](#)

In this manner, people can make and compare images made from *distinct* energy bands. However, in this manner, we may not study how the energy spectra vary *continuously* with positions. If we make a 3D cube of RA, DEC, Energy, we may see how the spectra vary *continuously* with positions.

Cas A is a popular source, which was observed with ASCA, Suzaku and XRISM (and Chandra, XMM etc). Another merit of our 3D tool is to display the ASCA, Suzaku and XRISM X-ray events together.

The following is a link to JUDO2, Suzaku and XRISM observation superposed.

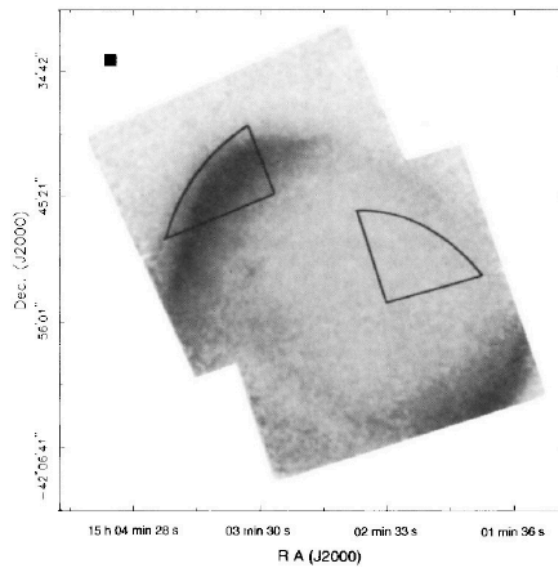
<https://darts.isas.jaxa.jp/app/judo2/?>

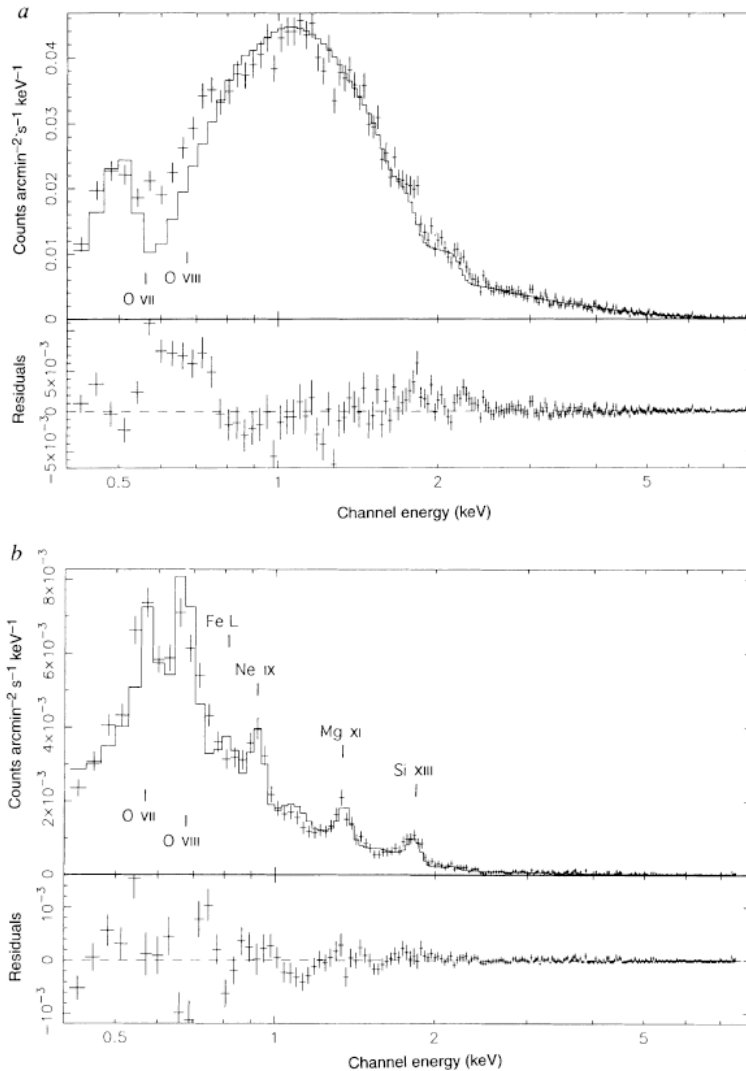
[center_lng=350.8584000¢er_lat=58.8113000&fov=3.201350&projection=AIT&coord=J2000&selectedLayer=XRISM_XTE](https://darts.isas.jaxa.jp/app/judo2/?center_lng=350.8584000¢er_lat=58.8113000&fov=3.201350&projection=AIT&coord=J2000&selectedLayer=XRISM_XTE)

In JUDO2, we may not superpose more than two instruments. Also, we may have to start UDON3 to see energy spectra. I would like to see how your 3D tools make the quick-analysis of Cas A easier; to see ASCA, Suzaku and XRISM data simultaneously, clarifying its spatial spectral variation easily.

SN1006 is another good example of indicating spatial variation. Here is a historical ASCA Nature paper which indicates that the supernova "rim" is bright in high energies due to synchrotron emission while the supernova "center" has low-temperature thermal emission with many emission lines.

[378255a0.pdf](#)





Here is JUDO2 link to SN1006, Suzaku and XRISM (xtend) observation.

[https://darts.isas.jaxa.jp/app/judo2/?](https://darts.isas.jaxa.jp/app/judo2/?center_lng=327.5794447¢er_lat=14.5617396&fov=1.966105&projection=AIT&coord=GAL&selectedLayer=XRISM_XTEND)

[center_lng=327.5794447¢er_lat=14.5617396&fov=1.966105&projection=AIT&coord=GAL&selectedLayer=XRISM_XTEND](https://darts.isas.jaxa.jp/app/judo2/?center_lng=327.5794447¢er_lat=14.5617396&fov=1.966105&projection=AIT&coord=GAL&selectedLayer=XRISM_XTEND)

Here is a list of the observation IDs for Cas A and SN1006 with ASCA, Suzaku and XRISM.

Please obtain the paquet files under <https://data.darts.isas.jaxa.jp/pub/app-data/udon3/>

I (Ken) would like to see how Cas A and SN1006 look like in your prototype 3D tools. Can we see spatial variation of the energy spectra easily?

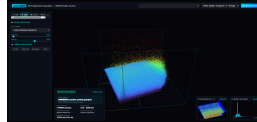
parquet file link:

- SN1006 XRISM (300080010) : <https://data.darts.isas.jaxa.jp/pub/app-data/udon3/xrism/300080010/>

- Cas A XRISM (000129000 and 000130000): <https://data.darts.isas.jaxa.jp/pub/app-data/udon3/xrism/000129000/> and <https://data.darts.isas.jaxa.jp/pub/app-data/udon3/xrism/000130000/>

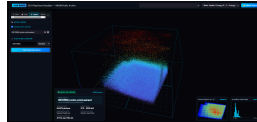
CasA_SN1006_OBSID

SN1006 xrism threeJS:

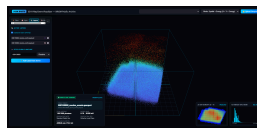


CasA XRISM threeJS:

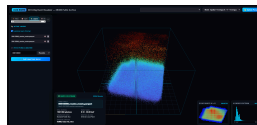
000129000



000130000



Both (000129000 and 000130000)



Ken: 2026-08-26

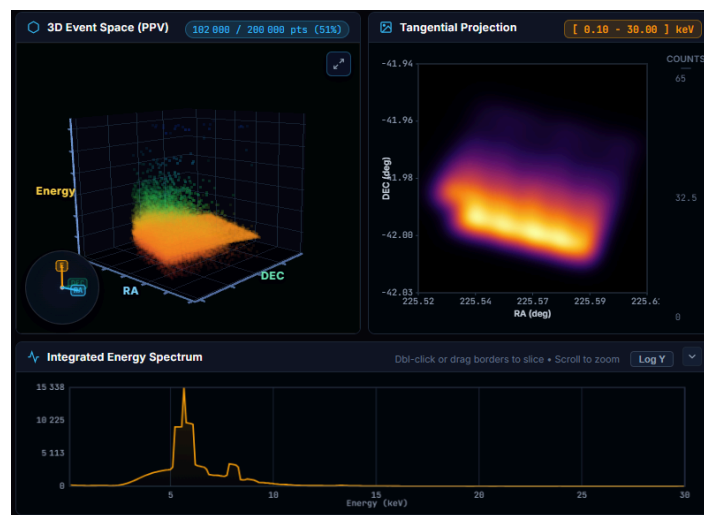
OK, you have already made good progress on Theme 2, so I will give actual tasks which I hope you to complete. If these (maybe not all) are completed, let's move to Theme 3. These tasks are both for Ramon and Theo, with using PyVista and Three.js-VTK.js, respectively, but I suppose you share some of the code.

1. Get central position on the sky and radius to display from user-input.
2. The position may be input either RA DEC in degree units or HHMMSS unit, or target name. To obtain the position from the target name, you may use sesame.
https://vizier.cds.unistra.fr/vizier/doc/sesame.htx?utm_source=chatgpt.com
3. Obtain list of the parquet files (ASCA, Suzaku, Hitomi and XRISM) from <https://data.darts.isas.jaxa.jp/pub/app-data/udon3/>
4. Properly convert the instrument-dependent PI to absolute Energy.
5. Display the events from these parquet files in 3D (RA, DEC, Energy). Users may rotate, zoom-in and zoom-out.
 - a. Initially, all the events from the eight instruments are shown (ASCA GIS, SIS, Suzaku XIS, Hitomi SXS, SXI, HXI, XRISM Resolve and Xtend). Users choose instruments to display.
 - b. Users may choose observation ID (paquet file) to display
6. Display "image" (compress the energy) on the RA DEC plane of Energy=constant. I am envisioning something like the following picture, which is an example of IDL (chatgpt gave me this). Note, in astronomical imaging, **the RA should increase toward the left-hand side.**
7. (added later) Display the "energy spectrum" (histogram of Energy) together. The energy spectrum may be displayed in a separate window from the "3D picture" and "image".
8. Display RA, DEC and Energy axis with labels.

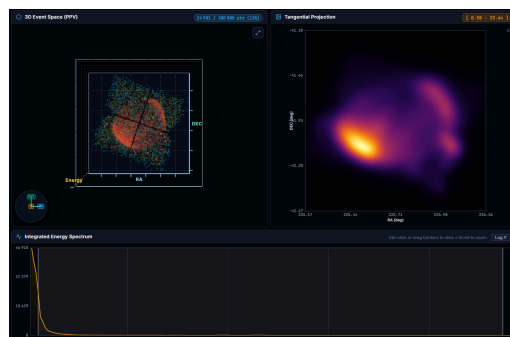
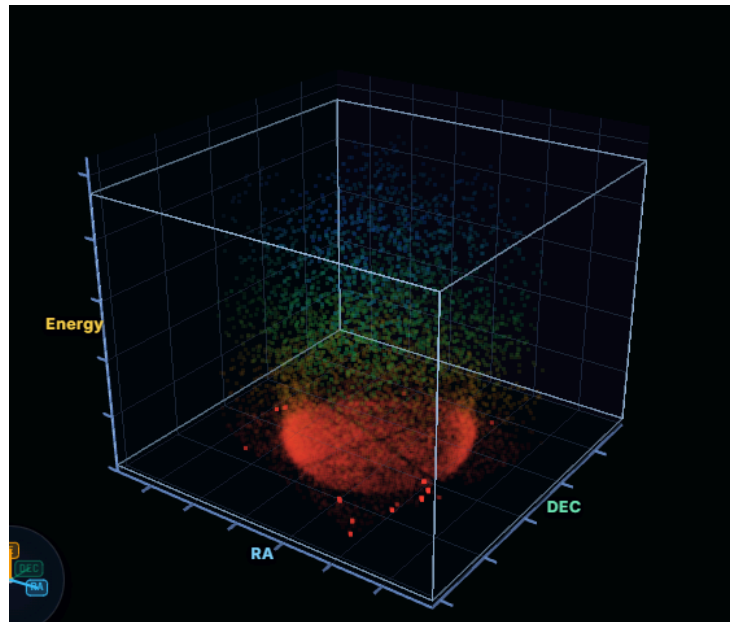
9. Put color on each event with RGB, such that the lowest energy is Red, and the highest energy Blue. The "image" will have similar colors.
 - a. User can choose the energies corresponding to Red, Green and Blue.
 - b. User can also to choose to have gradient / brightness for density of points with same energy
10. Show the same 3D events using voxels, where users may choose the voxel size.
 - a. Users may smooth the events or voxels to improve visibility. Image and Energy can have different smoothing parameter.

⇒ ? ⇒ maybe some features easier to see
11. Users can specify a energy band (specify the lower and higher boundaries), and move the band manually along the energy axis. Then the "image" is made only from that energy band.
12. Users can specify an area on RA and DEC (specify lower and higher values of RA and DEC independently). Then the energy spectrum is made only from that area.
13. Try your tool for Cas A and SN1006.
 - a. Reproduce the Cas A result above, indicating different images at different energy bands. ⇒ is it only with only XRISM file (000129000 or 000130000) ?? or several / different ?
 (Ken: the Cas A image I have shown above is from ASCA, from the paper in 1994 (!). After thirty years, combining ASCA, Suzaku and XRISM, we should be able to see the spectral variation of Cas A depending on the position in the SNR much more clearly. We will see.)
 - b. Confirm that SN1006 "shell" is bright above 4 keV, while the central part is dim in this energy range.
 (Ken) The image below seems to be only a part of the supernova remnant (SNR) shell. Can we see the entire SNR?
 I will show an example showing with JUDO2 (Suzaku and XRISM/xtend superposed):
 (Théo) maybe because I only use *resolve* file here ? (Ken: yes, that's right.)

with resolve file:



with xtend file:



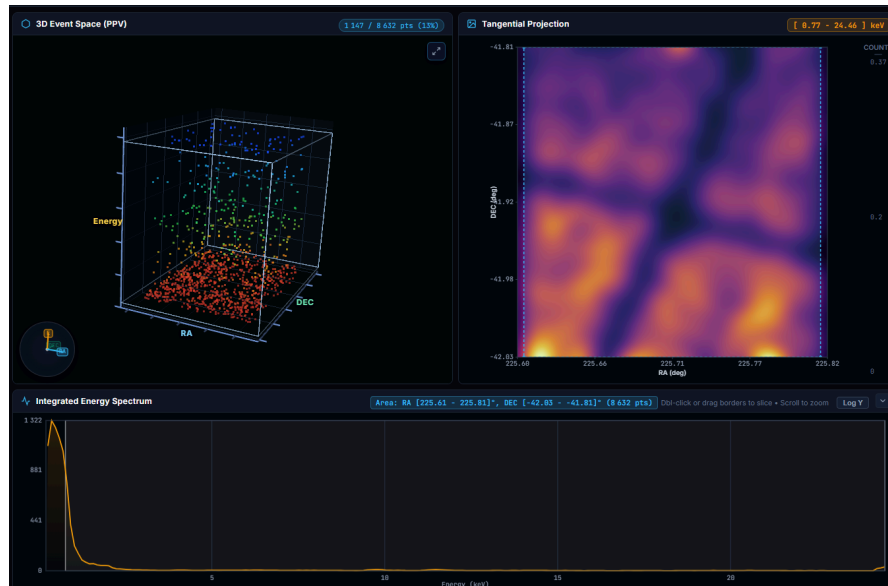
with judo2 (XRISM/xtend on Suzaku):

https://darts.isas.jaxa.jp/app/judo2?center_lng=225.7616782¢er_lat=-41.9489513&fov=2.637232&projection=AIT&coord=J2000&selectedLayer=&Top=XRISM_XTEND_IMAGE&Base=SUZAKU_IMAGE&TopAlpha=0.56&GraphicAlpha=1&consBoundary=false&reticle=false&grid=true&hpxgrid=false

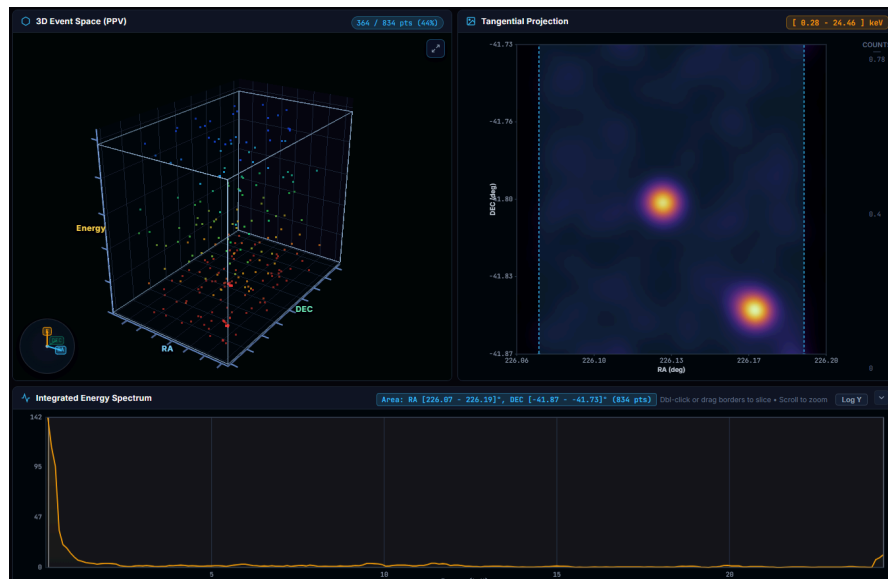
JUDO2: XRISM/resolve on XRISM/xtend, and the resolve spectrum. Note, resolve has only a very small field of view, while the spectral resolution is excellent. You see many prominent emission lines in the XRISM spectrum.

https://darts.isas.jaxa.jp/app/judo2?center_lng=225.7616782¢er_lat=-41.9489513&fov=2.402642&projection=AIT&coord=J2000&selectedLayer=&Top=XRISM_RESOLVE_IMAGE&Base=XRISM_XTEND_IMAGE&TopAlpha=0.56&GraphicAlpha=1&consBoundary=false&reticle=false&grid=true&hpxgrid=false

c. Choose particular RA and DEC area of Cas A and SN1006 to see the energy spectra in that area. For instance, if the central area of SN1006 is chosen, there will be a lot of emission lines, while no emission lines from the outer "shell". center of SN1006 (max number = ~1300): (Ken: show the energy spectrum in log-log scale. Can we recognize emission lines?)



outer shell (max number = ~140) : (Ken: outer-shell is bright above 4 keV, and should not show emission lines. Can we recognized that?)



14. (most important in 3D) Try to recognize the *continuous spatial variation* of the energy spectra in the 3D display for Cas A and SN1006. For both sources, we know the energy spectra is dependent on the source position, but that variation should not be disrupted, but should be continuous. The 3D representation hopefully help recognize such continuous spatial variation of the energy spectra.

⇒ maybe by growing ball size ?

15. Put the results on github, so that Nakahira-san and I can install, run and test.
16. If we are very successful, Ramon's desk-top tool may be public released from his github, and Theo's tool may be released from DARTS. We will see.

iacs01t4q_flt.fits: 623 267

